EXPERIMENT 19

Word Sense Disambiguation using Lesk Algorithm

AIM

To implement the simplified Lesk algorithm in Python to disambiguate the intended sense of a polysemous word based on context overlap.

PROCEDURE

1. Tokenize and clean the context sentence containing the ambiguous target word.
2. Retrieve candidate synsets for the target word from WordNet.
3. For each synset, compute the word overlap (intersection) between its definition/examples and the context sentence.
4. Select the synset having the maximum overlap score as the disambiguated sense.
5. Test and compare disambiguation across distinct contexts for words like 'bank' or 'bass'.

PROGRAM

```
import nltk
from nltk.corpus import wordnet as wn
from nltk.wsd import lesk

nltk.download('wordnet', quiet=True)
nltk.download('punkt', quiet=True)

sentences = [
    ("The financial bank deposited money into my checking account.", "bank"),
    ("The fishermen sat silently along the muddy river bank.", "bank"),
    ("He played a low bass note on his electric guitar.", "bass"),
    ("The chef prepared grilled sea bass with lemon herbs.", "bass")
]

print("--- Simplified Lesk Word Sense Disambiguation ---")
for sent, target_word in sentences:
    tokens = sent.split()
    best_synset = lesk(tokens, target_word)

    print(f"\nSentence: \"{sent}\"")
    print(f"Target Word: '{target_word}'")
    if best_synset:
        print(f"Disambiguated Synset: {best_synset.name()}")
        print(f"Definition: {best_synset.definition()}")
    else:
        print("Disambiguation could not resolve a synset.")
```

OUTPUT

```
> python3 experiment_19.py
--- Simplified Lesk Word Sense Disambiguation ---

Sentence: "The financial bank deposited money into my checking account."
Target Word: 'bank'
Disambiguated Synset: bank.n.05
Definition: a container (usually with a slot in the top) for keeping money at home

Sentence: "The fishermen sat silently along the muddy river bank."
Target Word: 'bank'
Disambiguated Synset: bank.n.01
Definition: sloping land (especially the slope beside a body of water)

Sentence: "He played a low bass note on his electric guitar."
Target Word: 'bass'
Disambiguated Synset: bass.n.07
Definition: the lowest part of the musical range

Sentence: "The chef prepared grilled sea bass with lemon herbs."
Target Word: 'bass'
Disambiguated Synset: sea_bass.n.01
Definition: the lean flesh of a saltwater fish of the family Serranidae
[Process completed with exit code 0]
```

RESULT

Thus, Word Sense Disambiguation using the Lesk algorithm was successfully implemented, correctly discerning senses based on contextual lexical overlap.